Semeton Balogun

Backend and systems engineer, moving into solutions architecture

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SUMMARY

Backend and systems engineer with six years in production: multi-tenant platforms for banks and enterprise SaaS, offline-first data for national health campaigns, and governance for AI coding agents. I start from the business goal: sizing infrastructure, designing deployments inside bank networks, and representing engineering with stakeholders. My current research is on agent governance: guardrails and repository memory for coding agents.

EXPERIENCE

eGov Foundation — Hub Software Engineer, West Africa (Contract) Jun 2026 – present

Bridge between country programme teams and platform engineering on DIGIT HCM, the open-source platform running Nigeria's national malaria campaigns.

- Diagnosed activity dates being stamped at sync time instead of on-device creation time, which locked field workers out of working days and skewed attendance-based pay; documented the root cause and fix requirement
- Contributed to the stock-reconciliation design for an ITN distribution pilot in an offline-first ledger
- Sizing infrastructure and support staffing for a 2027–29 programme across 11 states (~27M target population)

Seamless Technologies — Backend Engineer Nov 2024 – present

Recruitment platform and HRMS modules for a multi-tenant HR suite serving enterprise clients.

- Built the Kafka pipeline to the company data lake as a non-blocking side effect, so a mapping failure can't fail the primary write
- Implemented the two-way integration between the recruitment platform and the core HRMS, including SSO with company-access checks and service-to-service auth
- Led tenant-isolation and security hardening: SSRF-safe webhooks, company-scoped reads and writes, per-client mail credentials, GDPR self-service deletion
- Backend for AI-assisted hiring: CV parsing, candidate ranking, automated screening
- Volunteer facilitator on the Codeable programme, teaching software engineering to Special Olympics athletes and students at the Kirikiri correctional centre

Zercom Systems — Technical Lead Apr 2023 – Nov 2024

Led the rebuild of a per-client LMS into Tangerine365, one multi-tenant platform running on-premise at banks, as a self-serve cloud product, and inside Human Manager HRMS.

- Merged per-client codebases into one vanilla PHP 8 codebase with tenancy by subdomain or by ID, so a fix ships once instead of once per client
- Acted as solutions architect for client deployments: architecture, server sizing and specifications, and engineering's voice in stakeholder meetings
- Designed First Bank's three-tier on-premise deployment; deployed across IIS inside bank networks, Apache, DigitalOcean and AWS
- Wrote a SQL Server driver alongside the MySQL default, chosen at runtime, for banks that required it
- Consolidated the company's internal tools, staging and UAT onto one CyberPanel/LiteSpeed server with wildcard subdomains, cutting that hosting cost ~85%

ScholarshipIQ — Backend Engineer Jun 2022 – Apr 2023

- Cut API response times ~50% through query optimisation and indexing; introduced staging and branching discipline, cutting production errors by over 90%

Bincom Dev Center — Backend Developer (apprenticeship) Sep 2020 – Jul 2022

- Backend work in Node.js and PHP on ARM Learn for ARM Pensions and Social Lender; trained 5+ interns

SELECTED PROJECTS AND RESEARCH

Lawyering
Workspace for law firms and individual practitioners, built agent-first as a modular monolith of 19 Laravel modules with tenant scoping, encrypted file sharing and billing in naira and US dollars. In production since June 2026, with over 1,200 automated tests.

Repository memory for coding agents
Talk for PHP Talks Lagos and an experiment harness measuring whether a verified, human-gated decision log changes agent behaviour, including memory poisoning.

TrustCenta
Multi-tenant security platform wrapping OWASP ZAP, Gophish and CISO Assistant; rebuilt after a compromise and wrote the public post-mortem.

Agent Contract
npm CLI that checks every write an AI coding agent attempts against repository rules and blocks out-of-scope changes before they land.

Terminoxx360
ATM fleet monitoring at Access Bank; sole engineer on a multi-tenant rebuild with CBN regulatory uptime, automated escalation and an on-premise AI option.

EDUCATION

Miva Open University — Master of Information Technology, specialising in AI — In progress

University of Lagos — BSc Nursing Science, Distinction (joint-highest CGPA) — 2015 – 2021
Registered Nurse, Nursing and Midwifery Council of Nigeria, 2022.

SKILLS

Architecture: multi-tenancy, deployment design and infrastructure sizing, queues and event-driven workflows, offline-first sync, security hardening

Languages and frameworks: PHP (Laravel and vanilla), TypeScript, Node.js, React, Next.js, Livewire

Data and infrastructure: MySQL, PostgreSQL, SQL Server, Redis, Kafka, Elasticsearch, Docker, Linux, IIS, Apache, LiteSpeed, DigitalOcean, AWS

AI: Claude, OpenAI and Ollama integrations; agent-first development; governance for coding agents

WRITING

Engineering essays on Codeplified (semeton.substack.com), including *The Architecture I Didn't Design* and *Beyond Prompts: Repository Governance for AI Coding Agents*.